Fairness by Intervention: Client-Side Fairness Strategies and Server Auditing Protocols in Federated Learning

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Abstract

Federated learning (FL) can amplify demographic disparity under client-level heterogeneity, but fairness-targeted local interventions can also alter this behavior in non-trivial ways. We present a controlled empirical study of five client-side fairness interventions (I_1–I_5, legacy labels A_1–A_5) and five server-side auditing/stabilization protocols (P_1–P_5, legacy labels D_1–D_5). The evaluation uses UTKFace gender classification with `race_bin` as the protected attribute, ResNet18, 20 clients, 8 sampled clients per round, 30 rounds, three data regimes (IID, Dirichlet-α=0.5, strict non-IID), and three optimizers (SGD, AdamW, FedProx). We execute the full 189-configuration profile end-to-end (single seed: 42) and report descriptive statistics. The central finding is that optimizer choice dominates intervention-protocol outcomes: in intervention+protocol runs, AdamW reaches mean accuracy 0.7730, while SGD and FedProx collapse to 0.4545 and 0.4543, respectively. Among intervention-only settings, I_4 is most disruptive on average (ΔAcc $= -0.1500$ vs matched baselines). No protocol is uniformly best: P_4 attains the highest mean accuracy (0.5758) but also the largest mean equalized-odds gap (0.0879), whereas P_1 yields the lowest mean equalized-odds gap (0.0613) at lower mean accuracy (0.5529). Because strict non-IID contributes many low-utility runs, these means must be interpreted with regime context rather than as universal optimizer quality scores. The resulting picture is a strongly optimizer-conditional accuracy-fairness frontier, rather than a single dominant intervention or protocol policy.

1 Introduction

Federated Learning has emerged as a compelling privacy-preserving paradigm, yet several empirical and theoretical studies have shown that the standard FedAvg aggregation can *systematically disadvantage minority groups*. The root causes are diverse: statistical heterogeneity across clients, biased local data collection, and the implicit preference of FedAvg for clients with larger or more-frequent-class-representative datasets.

Prior work on fairness in FL broadly splits into two camps. *System-fairness* methods focus on balancing performance across clients. *Group-fairness* approaches target demographic criteria such as demographic parity and equal opportunity with respect to sensitive attributes. A third direction studies *client-side fairness interventions*, where clients modify local objectives, gradients, or representations to reduce group disparity. In this paper we adopt that language explicitly. There are no malicious Byzantine clients in our threat model; instead, the server protocols are auditing and stabilization mechanisms that characterize or bound intervention effects.

Contributions.

Preprint.

1. We introduce a taxonomy of five **client-side fairness interventions** (I_1 - I_5; legacy identifiers A_1 - A_5) of increasing complexity, from loss-reweighting to full disentangled debiasing networks.
2. We design five corresponding **server auditing protocols** (P_1 - P_5; legacy identifiers D_1 - D_5) and make explicit which intervention families are compatible with which protocols.
3. We provide a full 189-run benchmark over three data-heterogeneity regimes, three optimisers, one protected attribute (`race_bin`), and 30 communication rounds on UTKFace.
4. We provide a reproducible artifact bundle (full-run summaries, per-round logs, and figure/table source CSV files), plus optimizer-specific Pareto and convergence diagnostics.

2 Related Work

Federated optimisation. Canonical FL optimization is based on weighted averaging of client updates. Proximal variants add a regularisation term ($\mu\|\boldsymbol{\theta} - \boldsymbol{\theta}^t\|^2$) to reduce client drift under non-IID data. Adaptive server-side optimizers use momentum and variance adaptation to improve stability and convergence speed.

Non-IID heterogeneity. The Dirichlet distribution is widely used to simulate label skew. Prior empirical work reports that strong non-IID skew can degrade federated accuracy substantially (up to roughly 55% in extreme settings), while theoretical analyses provide convergence guarantees under bounded heterogeneity assumptions.

Group fairness in centralised learning. Centralized fairness literature established Equalized Odds and Equal Opportunity as core group-fairness criteria. Method families include adversarial debiasing, fairness-regularized training, and disentangled representations for multi-attribute fairness.

Group fairness in FL. FL fairness studies show that demographic gaps often widen as client heterogeneity increases. Proposed mitigation families include gradient-surgery constraints, fairness-aware server aggregation, reweighting of local objectives, and q-FedAvg-style resource allocation for client-level fairness.

This submission does *not* benchmark against FairFed, Ditto, AFL, or q-FedAvg directly; instead it provides a closed-world analysis of intervention-protocol interactions under a fixed FL stack.

Adversarial robustness in FL. Byzantine-resilient methods defend against malicious clients. Backdoor and targeted model-poisoning attacks have been demonstrated in federated training. Our work is complementary: we study *benign* interventions targeting demographic fairness rather than accuracy manipulation.

3 Problem Formulation

3.1 Federated Learning Setup

Consider a federation of K=20 clients $\{c_1, \ldots, c_K\}$ coordinated by a central server $\mathcal{S}$. Each client c_k holds a private dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^{n_k}$ where $\mathbf{x}_i \in \mathbb{R}^d$ is an image, $y_i \in \mathcal{Y}$ a task label, and $s_i \in \{0, 1\}$ is a protected attribute (in our experiments, `race_bin`). Let $f_{\boldsymbol{\theta}} : \mathbb{R}^d \to \mathcal{Y}$ be a ResNet18 parametrised by $\boldsymbol{\theta} \in \mathbb{R}^p$. The global objective is:

$$\min_{\boldsymbol{\theta}} \; F(\boldsymbol{\theta}) \;=\; \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\boldsymbol{\theta}), \qquad \mathcal{L}_k(\boldsymbol{\theta}) = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} \ell\big(f_{\boldsymbol{\theta}}(\mathbf{x}_i),\, y_i\big), \tag{1}$$

where $n = \sum_k n_k$ and ℓ is cross-entropy loss.

3.2 Fairness Metrics

For a global model $f_{\boldsymbol{\theta}}$ and protected attribute s, we measure:

$$\mathrm{DP}(\boldsymbol{\theta}) = \big|\mathbb{P}[\hat{y}=1 \mid s=0] - \mathbb{P}[\hat{y}=1 \mid s=1]\big|, \tag{2}$$

$$\Delta\mathrm{EO}(\boldsymbol{\theta}) = \big|\mathbb{P}[\hat{y}=1 \mid y=1, s=0] - \mathbb{P}[\hat{y}=1 \mid y=1, s=1]\big|. \tag{3}$$

Lower values indicate greater fairness. We additionally report *Equalised Odds* gap ($\mathrm{EO} = \max_{y\in\{0,1\}} \Delta\mathrm{EO}_y$). A primary evaluation criterion is the *accuracy-fairness Pareto front*: pairs $(\mathrm{Acc}, \Delta\mathrm{EO})$ across interventions and protocols.

3.3 Data Partitioning Regimes

IID. Each client samples uniformly from the global pool; identical marginal distributions.

Dir(α=0.5). Label proportions per client are sampled from a Dirichlet distribution with concentration α=0.5, producing moderate but realistic label skew.

Strictly non-IID. Each client receives samples from at most two class labels, maximising label skew.

4 Methodology

4.1 Client-Side Fairness Interventions

All five interventions are *client-side only*: the default server is unchanged unless a protocol is explicitly enabled. We use four families: loss-level (I_1), gradient-level (I_2), constraint-level (I_3), and representation-level (I_4, I_5). For reproducibility with exported artifacts, we keep legacy labels A_1–A_5 in parentheses.

I_1 (legacy A_1): Loss Reweighting (LR). Client c_k upweights samples from the underrepresented protected group:

$$\hat{\mathcal{L}}_k^{\mathrm{LR}}(\boldsymbol{\theta}) = \frac{1}{n_k} \sum_{i\in\mathcal{D}_k} w(s_i)\, \ell\big(f_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i\big), \quad w(s) = \frac{n_k}{2\, n_{k,s}}, \tag{4}$$

where $n_{k,s}$ is the count of group-s samples in $\mathcal{D}_k$.

I_2 (legacy A_2): Gradient Projection (GP). Following, the local gradient is decomposed and the component that conflicts with the fairness gradient is suppressed:

$$\tilde{g}_k = g_k - \frac{\langle g_k,\, g_k^{\mathrm{fair}}\rangle}{\|g_k^{\mathrm{fair}}\|^2} g_k^{\mathrm{fair}}, \tag{5}$$

where $g_k^{\mathrm{fair}} = \nabla_{\boldsymbol{\theta}}\, \Delta\mathrm{EO}(\boldsymbol{\theta}; \mathcal{D}_k)$.

I_3 (legacy A_3): Fairness-Constrained Local Optimisation (FCLO). Local training minimises task loss subject to an approximate fairness constraint via augmented Lagrangian:

$$\min_{\boldsymbol{\theta}}\ \mathcal{L}_k(\boldsymbol{\theta}) + \lambda \max\big(0,\ \Delta\mathrm{EO}(\boldsymbol{\theta}; \mathcal{D}_k) - \epsilon\big)^2, \tag{6}$$

where ϵ=0.05 is the fairness tolerance and λ is a penalty coefficient.

I_4 (legacy A_4): Representation Disentanglement (RD). Each client trains an auxiliary encoder $e_\phi : \mathbb{R}^d \to \mathbb{R}^h$ that is *orthogonal* to s via gradient reversal:

$$\min_{\boldsymbol{\theta},\phi}\ \mathcal{L}_k(\boldsymbol{\theta}) + \mu_1 \mathcal{L}_k^{\mathrm{adv}}(\phi, s) - \mu_2 \mathcal{L}_k^{\mathrm{adv}}(\phi, y), \tag{7}$$

where $\mathcal{L}_k^{\mathrm{adv}}$ denotes the discriminator loss. Only $\boldsymbol{\theta}$ (backbone + head) is transmitted to the server.

$\mathbf{I}_5$ (legacy $\mathbf{A}_5$): Full Bi-level Debiasing (FBD). Extends A_4 to a bi-level optimisation, simultaneously learning a protected-attribute discriminator and imposing Eq. (6):

$$\min_{\boldsymbol{\theta}} \max_{\phi} \; \mathcal{L}_k(\boldsymbol{\theta}) - \beta_1 \, \mathcal{L}_k^{\text{disc}}(\phi; \boldsymbol{\theta}) + \beta_2 \max\big(0, \; \Delta\text{EO} - \epsilon\big)^2. \tag{8}$$

4.2 Server Auditing and Stabilization Protocols

$\mathbf{P}_1$ (legacy $\mathbf{D}_1$): Fairness-Audited Aggregation (FAA). After each round, the server estimates ΔEO on a held-out audit shard and flags rounds where fairness improvement exceeds a threshold τ. Flagged rounds trigger re-weighting of client contributions proportional to their accuracy contribution.

$\mathbf{P}_2$ (legacy $\mathbf{D}_2$): Gradient Similarity Filtering (GSF). Cosine similarity between each client's gradient and the fair-direction proxy is monitored. Updates with similarity above threshold γ are clipped: $\tilde{g}_k \leftarrow \text{clip}(g_k, \gamma)$.

$\mathbf{P}_3$ (legacy $\mathbf{D}_3$): Proximal Fairness Bounding (PFB). Combines FedProx with a server-side Lagrangian that *limits* global fairness change per round, preventing runaway debiasing that degrades accuracy.

$\mathbf{P}_4$ (legacy $\mathbf{D}_4$): Representation Fingerprinting (RF). Detects I_4 by monitoring mutual information between transmitted representation $e_\phi(\mathbf{x})$ and s, estimated via MINE. Clients exhibiting anomalously low MI(e; s) receive dampened aggregation weights.

$\mathbf{P}_5$ (legacy $\mathbf{D}_5$): Stability Testing (ST). Full detection protocol for I_5: the server maintains a shadow model trained without fairness interventions. Divergence between global and shadow gradients triggers alert and selective rollback.

4.3 Compatibility Design and Evaluation Matrix

The intervention-protocol grid is intentionally asymmetric. I_1–I_3 are loss/gradient/constraint interventions whose signatures are observable in gradient-space statistics, so they are evaluated with P_1–P_3. I_4–I_5 are representation-level interventions; P_4–P_5 require representation-specific probes and are therefore evaluated only for that family. P_3 is shared across both families because it operates at update-space stabilization without requiring representation internals.

Table 1: Loss/gradient/constraint family: intervention-only vs P_1–P_3. Each cell reports mean (Acc / EO gap), aggregated over split$\times$optimizer.

	Intervention only	$\mathbf{P}_1$	$\mathbf{P}_2$	$\mathbf{P}_3$
$\mathbf{I}_1$ ($\mathbf{A}_1$)	0.759 / 0.028	0.539 / 0.059	0.566 / 0.060	0.569 / 0.073
$\mathbf{I}_2$ ($\mathbf{A}_2$)	0.769 / 0.043	0.555 / 0.059	0.562 / 0.070	0.557 / 0.063
$\mathbf{I}_3$ ($\mathbf{A}_3$)	0.722 / 0.034	0.564 / 0.066	0.565 / 0.071	0.549 / 0.069

Table 2: Representation family: intervention-only vs P_3–P_5. P_3 is the shared update-space stabilizer across families.

	Intervention only	$\mathbf{P}_3$	$\mathbf{P}_4$	$\mathbf{P}_5$
$\mathbf{I}_4$ ($\mathbf{A}_4$)	0.628 / 0.043	0.567 / 0.105	0.590 / 0.082	0.561 / 0.070
$\mathbf{I}_5$ ($\mathbf{A}_5$)	0.683 / 0.043	0.568 / 0.063	0.562 / 0.094	0.535 / 0.072

5 Experimental Setup

Dataset. UTKFace is parsed from filenames that encode age, gender, race, and date. The prediction target is binary gender, while fairness is evaluated on a binary protected attribute derived from race labels: value 0 for race classes {0,1}, and value 1 otherwise. Images are resized to 128×128 and normalised with ImageNet statistics.

Model. We use ResNet18 with a binary classification head. When available, ImageNet pretrained weights initialise the backbone once and are reused across runs; otherwise random initialisation is used.

Federation. One server, $K = 20$ clients. Data is partitioned using the three regimes described in Section 3. Each round samples 8 clients (partial participation).

Training. The full profile executes 30 communication rounds with batch size 64, 4 data-loading workers, and 1 local epoch per selected client. Three optimizer configurations are compared: (i) **SGD/FedAvg**: SGD with momentum 0.9, learning rate 0.01, weight decay 10^{-4}; (ii) **AdamW/FedAvg**: AdamW with learning rate 10^{-3}, weight decay 10^{-4}; (iii) **FedProx**: SGD with learning rate 0.01 and proximal coefficient $\mu = 10^{-3}$. Automatic mixed precision is enabled.

Evaluation protocol. After each round, the global model is evaluated on a centralised held-out validation split (15% of the parsed dataset). We report Top-1 accuracy, DP, and equalized-odds gap (EO) (Section 3.2). The submitted full run uses seed 42 and executes all $1 + 5 + 15 = 21$ conditions across 3 splits and 3 optimizers, totaling 189 runs. Because the run is single-seed, this paper reports descriptive statistics only (no confidence intervals).

Benchmark scope and exclusions. This submission evaluates intervention-protocol interactions within a fixed FL pipeline. We do not include external fairness baselines (FairFed, Ditto, AFL, q-FedAvg) or a centralized oracle in this draft; these are explicit follow-up experiments, not hidden omissions.

6 Results

The full profile completed all 189 runs (9 baseline, 45 intervention-only, 135 intervention+protocol). Per-run duration was 210–281 seconds (mean 243.34 seconds), for 12.78 cumulative compute hours.

6.1 Vanilla FL Baseline

Table 3: Baseline runs (condition `vanilla`) for each split and optimizer.

Split	Optimizer	Accuracy	DP gap	EO gap
iid	sgd	0.9181	0.0230	0.0607
iid	adamw	0.9297	0.0400	0.0805
iid	fedprox	0.9215	0.0234	0.0532
dirichlet_0_5	sgd	0.8563	0.0739	0.1368
dirichlet_0_5	adamw	0.8411	0.0021	0.0171
dirichlet_0_5	fedprox	0.7170	0.0257	0.0338
strict_non_iid	sgd	0.4824	0.0000	0.0000
strict_non_iid	adamw	0.8540	0.0106	0.0230
strict_non_iid	fedprox	0.4824	0.0000	0.0000

Baseline performance already shows strong optimizer dependence under heterogeneity: AdamW remains competitive across all splits, whereas SGD/FedProx collapse on strict non-IID.

6.2 Intervention-Only Behavior

I_4 is the most disruptive intervention in mean accuracy, while I_1/I_2 retain higher utility. Fairness changes should be interpreted jointly with utility, since low EO can coincide with near-degenerate accuracy on hard splits.

6.3 Protocol Effects and Metric Disagreement

No protocol dominates globally. P_4 provides the highest mean accuracy but at the highest fairness gap; P_1 provides the best mean EO gap but with the largest utility degradation. DP and EO are

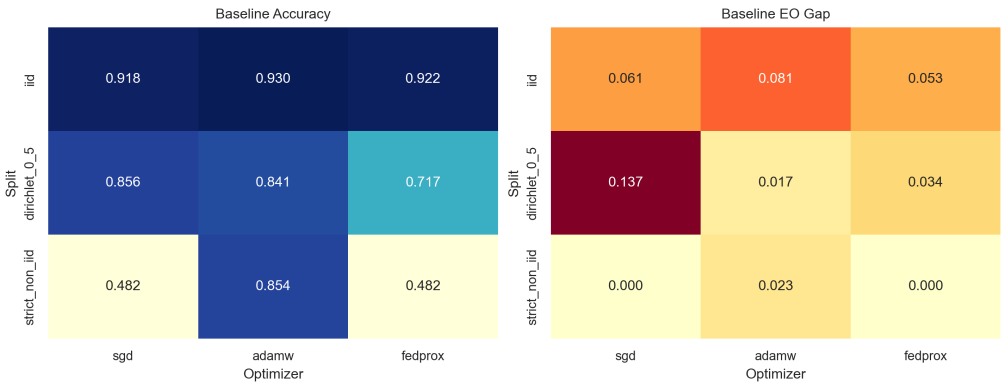

Figure 1: Baseline heatmaps for accuracy (left) and EO gap (right): AdamW stays stable across splits, while SGD/FedProx fail on strict non-IID.

Table 4: Intervention-only averages across split×optimizer.

Intervention	Mean Acc	Mean DP	Mean EO	ΔAcc vs matched baseline
I_1 (A_1)	0.7592	0.0155	0.0276	-0.0188
I_2 (A_2)	0.7688	0.0238	0.0431	-0.0093
I_3 (A_3)	0.7222	0.0151	0.0339	-0.0558
I_4 (A_4)	0.6281	0.0279	0.0433	-0.1500
I_5 (A_5)	0.6833	0.0227	0.0428	-0.0948

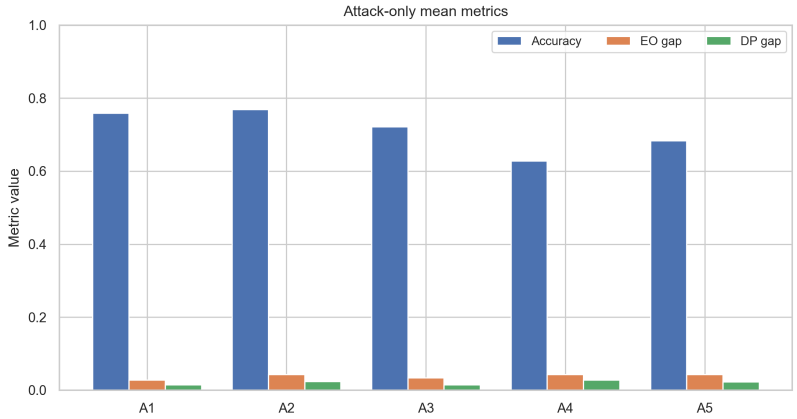

Figure 2: Intervention-only means by family: representation-level interventions (I_4, I_5) incur the largest utility cost.

Table 5: Protocol-wise averages over all compatible interventions, splits, and optimizers.

Protocol	Mean Acc	Mean DP	Mean EO	ΔAcc vs intervention-only
P_1 (D_1)	0.5529	0.0472	0.0613	-0.1971
P_2 (D_2)	0.5643	0.0500	0.0671	-0.1858
P_3 (D_3)	0.5619	0.0546	0.0744	-0.1504
P_4 (D_4)	0.5758	0.0682	0.0879	-0.0799
P_5 (D_5)	0.5481	0.0511	0.0711	-0.1076

not interchangeable in this matrix: in 15 of 45 intervention×split×optimizer cells, the protocol that minimizes DP is different from the one that minimizes EO.

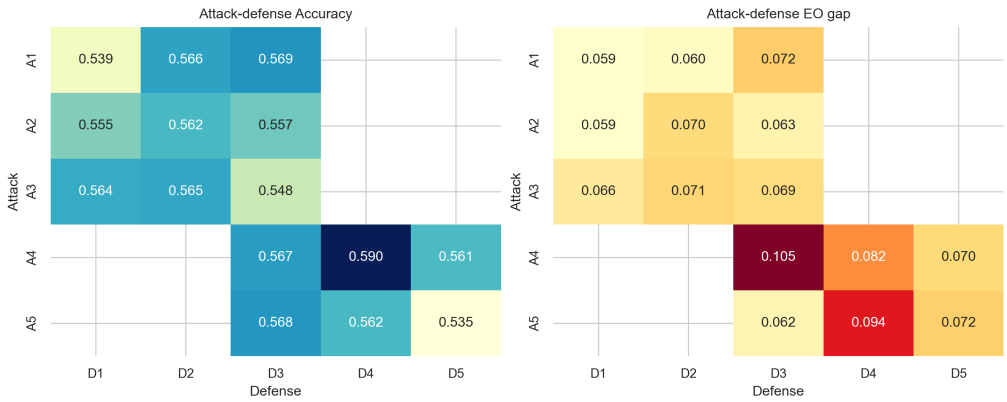

Figure 3: Intervention-protocol matrix heatmaps: P_4 often maximizes accuracy but tends to worsen EO relative to P_1.

Table 6: Optimizer means by experimental category.

Category	Optimizer	Mean Acc	Mean EO
baseline	adamw	0.8749	0.0402
baseline	fedprox	0.7070	0.0290
baseline	sgd	0.7523	0.0658
intervention_only	adamw	0.7687	0.0511
intervention_only	fedprox	0.6931	0.0393
intervention_only	sgd	0.6752	0.0240
intervention_protocol	adamw	0.7730	0.0579
intervention_protocol	fedprox	0.4543	0.0784
intervention_protocol	sgd	0.4545	0.0788

6.4 Optimizer-Conditional Robustness and Degenerate Regimes

The dominant empirical pattern is optimizer-conditional robustness: in intervention+protocol runs, AdamW substantially outperforms SGD/FedProx. Therefore, conclusions about "effective protocols" are invalid unless optimizer is explicitly controlled.

Table 7: Effect of removing degenerate runs (Acc ≤ 0.55) from intervention+protocol aggregates.

Subset	Optimizer	Mean Acc	Mean EO	N runs
All runs	adamw	0.7730	0.0579	45
All runs	sgd	0.4545	0.0788	45
All runs	fedprox	0.4543	0.0784	45
Acc $>$ 0.55 only	adamw	0.8576	0.0676	34
Acc $>$ 0.55 only	sgd	–	–	0
Acc $>$ 0.55 only	fedprox	–	–	0

The low means for SGD/FedProx in Table 6 are therefore largely driven by strict non-IID collapse rather than competitive low-accuracy equilibria.

6.5 Pareto Frontier, Round Dynamics, and Split Diagnostics

The best-accuracy run is A_2+D_2 under IID with AdamW (accuracy 0.9384, EO 0.0426). Relative to the matched IID+AdamW baseline (0.9297), this is a +0.0087 gain, suggesting that some intervention-protocol pairings can act as regularizers rather than purely harmful perturbations. The largest EO outlier is A_4+D_3 under strict non-IID with AdamW (accuracy 0.6138, EO 0.4126), illustrating that high utility and extreme fairness disparity can co-occur.

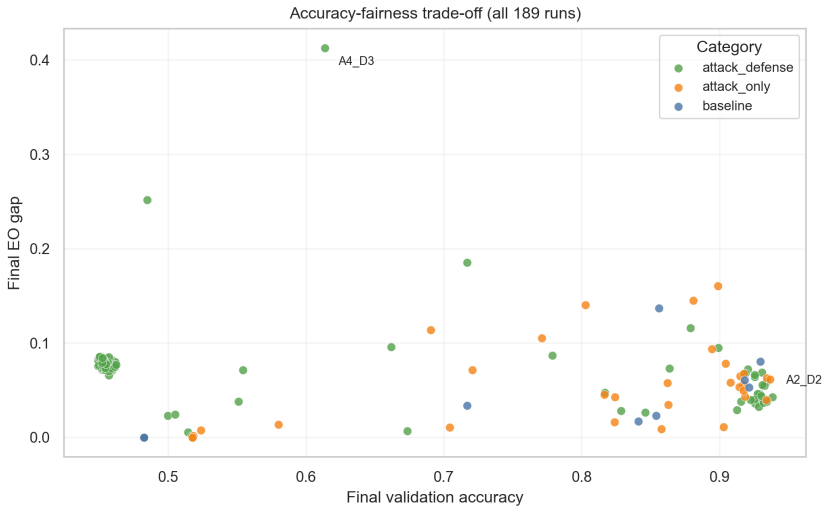

Figure 4: Accuracy-EO scatter across all 189 runs: the top-accuracy point and largest-EO outlier are both visible and should be interpreted jointly.

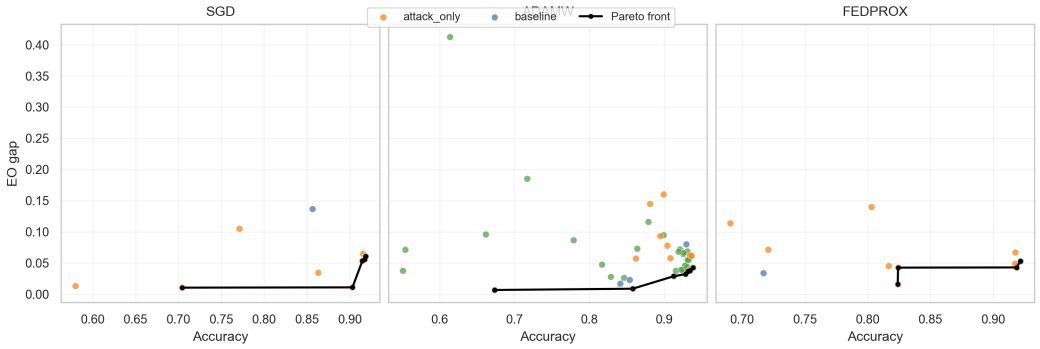

Figure 5: Optimizer-specific Pareto fronts after excluding degenerate runs (Acc ≤ 0.55): AdamW forms the strongest frontier across most of the practical operating range.

Table 8: Convergence proxy by split/optimizer: fraction of runs with $|\Delta \mathrm{Acc}_{26\to30}| < 0.01$.

Split	**SGD**	**AdamW**	**FedProx**
iid	0.905	0.952	0.952
dirichlet_0_5	0.762	0.286	0.619
strict_non_iid	0.857	0.429	0.905

Split statistics explain this behavior: IID is tightly balanced (samples/client 1007–1008), Dirichlet exhibits severe sample skew (64–2295), and strict non-IID has extreme label/protected rates (0 to 1). Under strict non-IID, 54/63 runs fall in 0.45–0.55 accuracy, and 23/63 runs report EO gap 0.0, indicating frequent near-degenerate operating points.

7 Discussion

Effect of heterogeneity. Performance degrades with split severity. Across all runs, mean accuracy is 0.6874 (IID), 0.6413 (Dirichlet), and 0.4926 (strict non-IID). The strict non-IID setting frequently enters near-degenerate regimes, which can artificially deflate fairness gaps without meaningful utility.

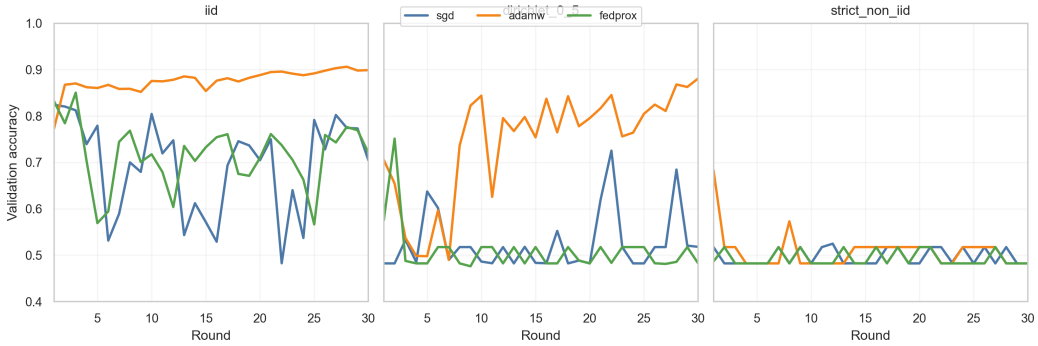

Figure 6: Round-wise dynamics for I_4 (A_4): collapse behavior depends strongly on split and optimizer, and several non-IID cells remain non-stationary at 30 rounds.

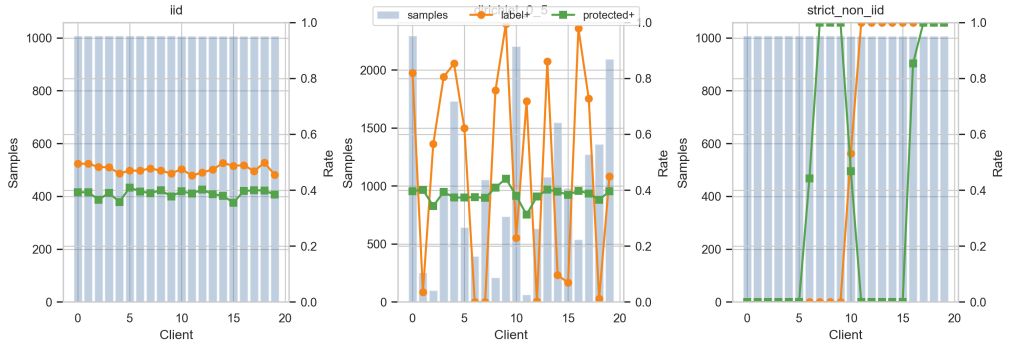

Figure 7: Per-client sample and rate diagnostics: strict non-IID reaches extreme 0/1 label and protected-group regimes, explaining frequent degenerate operating points.

Intervention complexity at scale. I_4 and I_5 are the most utility-disruptive interventions on average, with matched-baseline drops of -0.1500 and -0.0948. In contrast, I_1 and I_2 preserve substantially more accuracy. This indicates that stronger intervention complexity does not translate into uniformly better accuracy-fairness operating points.

Protocol effectiveness. No protocol is globally dominant. P_4 yields the best mean accuracy (0.5758) but the worst mean EO gap (0.0879), while P_1 yields the best mean EO gap (0.0613) at lower mean accuracy (0.5529). Hence, protocol choice should be treated as a policy decision over a utility-fairness frontier, not as a one-dimensional optimization.

Optimiser sensitivity. Optimizer choice is the strongest factor in this benchmark. In intervention+protocol conditions, AdamW reaches mean accuracy 0.7730, whereas SGD and FedProx remain near 0.454. The degenerate-filtered analysis (Table 7) shows that this gap is largely a collapse-vs-non-collapse phenomenon. Therefore, any claim about intervention or protocol effectiveness must be conditioned on optimizer.

Mechanistic interpretation (hypothesis-level). The optimizer gap is consistent with adaptive-step-size stabilization: AdamW appears more resilient to intervention-induced gradient distortions than SGD/FedProx in high-heterogeneity regimes. Our current evidence is empirical rather than causal; targeted ablations separating adaptation, momentum, and weight decay are required before making mechanistic claims.

Framing boundary. This paper analyzes *benign unilateral fairness interventions*, not malicious Byzantine attacks. We keep legacy A/D labels for reproducibility with artifacts, but the appropriate interpretation is intervention strategies and server auditing/stabilization protocols.

Limitations.

1. The study is single-dataset (UTKFace); cross-dataset generalization remains unverified.
2. The submitted full profile uses one seed, so variance estimation is limited and should be extended in future work.
3. External fairness FL baselines (FairFed, Ditto, AFL, q-FedAvg) are not yet included, so cross-method ranking claims are out of scope for this draft.
4. No centralized oracle baseline is reported, so FL-vs-centralized fairness gaps are not isolated.
5. The protected attribute is a binary race proxy (`race_bin`), which does not capture intersectional fairness structure.
6. Hyperparameter sensitivity is not yet mapped (e.g., strength, clipping, and threshold schedules), so rank ordering may shift outside the tested setting.
7. Strict non-IID frequently produces near-degenerate accuracy, complicating interpretation of very low fairness gaps.
8. We do not model client incentives formally, so when and why clients would adopt these interventions in practice remains open.

8 Conclusion

We presented a controlled empirical study of client-side fairness interventions and server auditing protocols in federated learning. Our full 189-configuration benchmark over three split regimes and three optimizers shows that optimizer choice dominates intervention-protocol outcomes: AdamW remains robust under intervention pressure, while SGD/FedProx often collapse in intervention+protocol settings. We also show that no protocol is universally optimal, with clear trade-offs between utility and fairness metrics. These findings motivate optimizer-aware fairness governance in FL. The next experimental milestone is a multi-seed, external-baseline, and sensitivity-complete benchmark to convert these descriptive trends into statistically robust comparative conclusions.

A Extended Experimental Details

A.1 UTKFace Preprocessing

UTKFace images are loaded at variable native resolution, resized to 128×128 via bilinear interpolation, and normalised channel-wise: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$. The pipeline uses deterministic resize+normalize transforms.

A.2 Client Partitioning Details

Table 9: Measured client-level data statistics by split (seed 42).

Split	Samples/client	Sample range	Label+ range	Protected+ range
iid	1007.5 ± 0.5	1007–1008	0.4528–0.4985	0.3555–0.4097
dirichlet_0_5	1007.5 ± 676.45	64–2295	0.0000–0.9946	0.3125–0.4417
strict_non_iid	1007.5 ± 0.5	1007–1008	0.0000–1.0000	0.0000–1.0000

A.3 Full Hyperparameter Table

A.4 Artifact Files

The run exports all report-facing artifacts as CSV/JSON:

- `all_runs_final.csv`, `all_rounds.csv`
- `table2_baseline.csv`
- `fig1_fig3_fig5_attack_only.csv`
- `table1_fig4_attack_defense.csv`
- `fig2_data_distribution.csv`

Table 10: Complete hyperparameter grid used in all experiments.

Parameter	**Value(s)**	**Notes**
Communication rounds T	30	Full profile
Clients K	20	Total clients
Clients per round	8	Partial participation
Local epochs E	1	Per selected client
Batch size	64	All clients
`num_workers`	4	DataLoader
Validation split	0.15	Hold-out validation
SGD learning rate	10^{-2}	Momentum 0.9
SGD momentum	0.9	
SGD weight decay	10^{-4}	
AdamW learning rate	10^{-3}	
AdamW (β_1, β_2)	$(0.9, 0.999)$	
AdamW weight decay	10^{-4}	
FedProx learning rate	10^{-2}	
FedProx μ	10^{-3}	Proximal coefficient
Intervention strength	0.25	Shared coefficient
Protocol strength	0.25	Shared coefficient
Checkpoint period	Every 5 rounds	Resume enabled
Seed list	$\{42\}$	Single-seed submission run
AMP	Enabled	Mixed precision training

- `fig6_scatter_source.csv`

These files are sufficient to regenerate all tables and figures in this submission.